THE TRANSPORT ACTIVITY OF PTCH1 IN HEDGEHOG SIGNALING REGULATION AND IN CHEMOTHERAPY RESISTANCE OF CANCER CELLS

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ABSTRACT

PTCH1 is a 12 transmembrane helices protein from the RND efflux pumps which regulates the GPCR Smoothened and the Hedgehog signaling by transporting cholesterol.

PTCH1 is overexpressed in many recurrent and metastatic cancers. We showed that PTCH1pumps chemotherapeutic agents such as doxorubicin out of cancer cells using the proton motive force like the other RND transporters, and contributes to chemotherapy resistance of several cancer cell types. We identified a small molecule which inhibits the doxorubicin efflux activity of PTCH1 and enhances its cytotoxicity in different cancer cell lines endogenously overexpressing PTCH1 and thereby mitigates the resistance of these cancer cells to doxorubicin. This molecule, panicein A hydroquinone (PAH) was first extracted from marine sponges and then chemically synthesized to develop a drug candidate. We recently showed that an optimized derivative of PAH enhances the efficacy of kinase inhibitors such as vemurafenib against melanoma cells resistant to the treatment in cellulo and in vivo. Data obtained from microscale thermophoresis, in silico docking and molecular dynamics, and using a fluorescent derivative of PAH strongly support that this drug efflux inhibitor interacts directly with PTCH1 and allowed proposing its inhibition mechanism.

Altogether, our data suggest that the use of inhibitors of PTCH1 drug efflux in combination with chemotherapy could be a promising therapeutic option to improve chemotherapy efficiency against cancer cells expressing PTCH1.

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